

Multiple Choice (10 marks)

1. A major difference between cancer cells and normal cells is that
 - (a) Cancer cells have no telomere strands
 - (b) Normal cells become cancerous when their telomere strands are used up
 - (c) Cancer cells have no upper limit on cell division
 - (d) Normal cells are protected from cancer when they have telomerase
2. All other things being equal, which of the following men is likely to be the happiest in retirement?
 - (a) Gary who is still married
 - (b) Leo who never married
 - (c) Josh who is divorced
 - (d) Caleb who is remarried
3. Which level of personality description appears to show the most change with age?
 - (a) Traits
 - (b) Personal concerns
 - (c) Coping strategies
 - (d) Identity
4. The parvoviruses have a rather unique genome; what form does it take?
 - (a) Circular ds DNA
 - (b) Small linear ss DNA
 - (c) ds DNA genome
 - (d) RNA genome with RT
5. What is the most successful and most widely used 'travel' vaccine for international visitors?
 - (a) Influenza
 - (b) MMR
 - (c) Yellow Fever
 - (d) Rabies

True / False (10 marks)

Answer True or False

6. True or False: DNA damage and repair theory claims that senescence is primarily caused by radiation from anti-oxidants.
7. True or False: The prepuce, commonly known as the foreskin, is removed during male circumcision.
8. True or False: In X-linked dominant inheritance, males transmit the trait only to their female offspring.
9. True or False: Adaptive immunity responds quickly to foreign proteins, similar to the innate immune response.
10. True or False: Public health ethics is primarily focused on the health and well-being of populations rather than individual patients.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the implications and applications of the prone position in patient care, including its benefits, risks, and scenarios where it is utilized in health sciences.
12. Discuss the limitations of next generation sequencing (NGS) in health sciences, providing examples of scenarios where it may not be applicable or effective.

End of Paper